

Section 87 — Master Matching And Matroids Mixed Contest

Contest contract

Six new problems combine threshold matching, blossom modeling, cut-tree aggregation, matroid intersection, and directed arborescence contraction.

Problem	Main idea
A	binary search plus bipartite feasibility
B	unmatched vertices from general matching
C	all pair cuts represented by tree paths
D	graphic/partition matroid intersection
E	minimum incoming edges and cycle contraction
F	common spanning tree in two graphic matroids

Recognition map

Ask first what a feasible object is, not which named algorithm seems nearby.

- A bijection between two sides suggests bipartite matching or assignment.
- Disjoint pairs in one unrestricted vertex set require general matching.
- Many pair minimum-cut queries on one undirected graph suggest a cut tree.
- A largest set satisfying two hereditary exchange systems suggests matroid intersection; identify both independence oracles explicitly.
- One incoming edge per non-root vertex in a directed graph is not ordinary MST. Directed cycles lead to arborescence contraction.

Two representative derivations

For bottleneck assignment, fix a threshold T and keep only edges of cost at most T . A perfect matching is now exactly a certificate that maximum chosen cost is at most T . Increasing T only adds edges, so feasibility is monotone and binary search becomes justified. This “optimize the maximum by deciding a threshold” pattern is reusable far beyond matching.

For directed arborescence, first choose the cheapest incoming edge of every non-root vertex. If these choices contain no directed cycle, they already form the optimum. If a cycle appears, every arborescence must replace at least one of its chosen incoming edges. Add all chosen costs, contract the cycle, and reduce each

entering edge by the chosen cost of the vertex it enters. The reduced value is exactly the extra price of choosing that entry instead.

Contest proof checklist

For each solution, answer all four questions before coding:

1. What mathematical objects correspond one-to-one with valid outputs?
2. Which invariant survives an augmentation or contraction?
3. What certifies that no better object remains?
4. Do signed weights, parallel edges, loops, or an impossible target alter the reduction?

Tiny exhaustive oracles are especially valuable here: enumerate all assignments, matchings, or subsets for small instances and compare objective values, not only validity.

Exercises

A–F form the complete mixed contest.